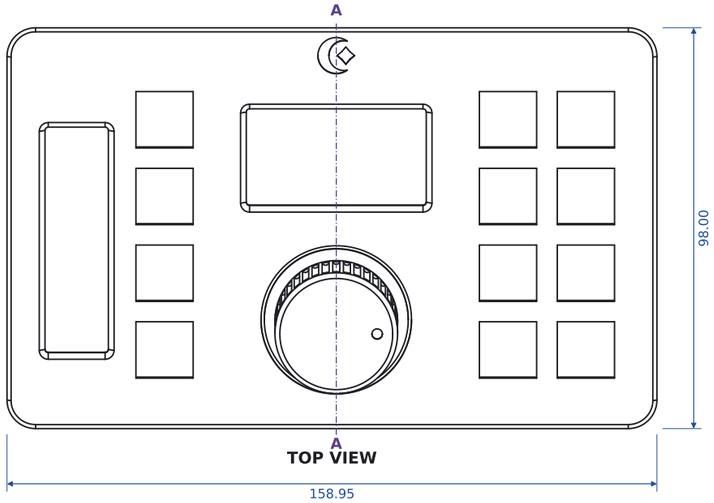
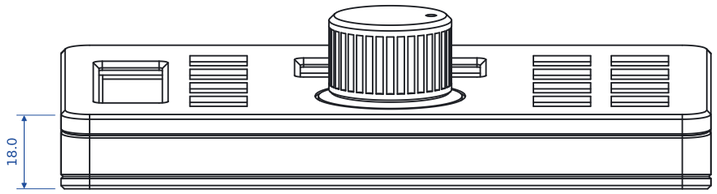
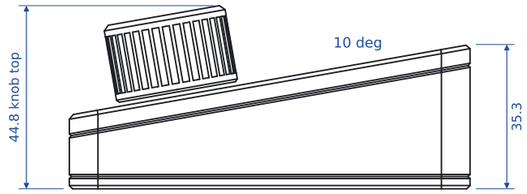


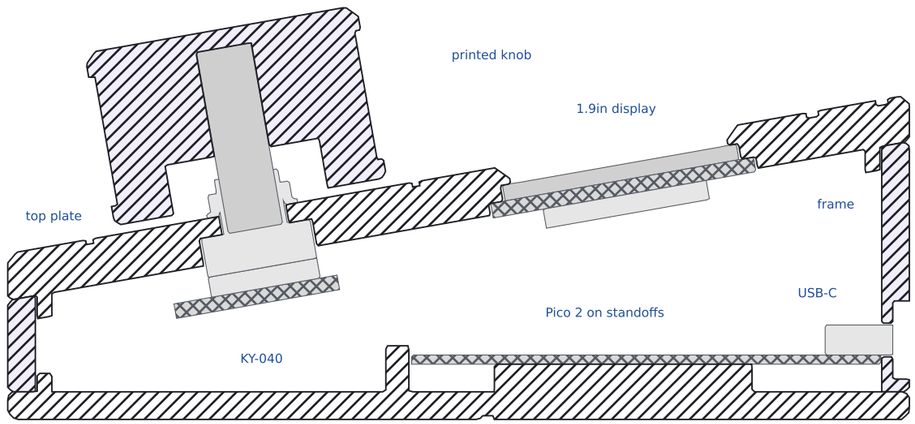
TOP VIEW



FRONT VIEW (user side)



RIGHT SIDE VIEW



SECTION A-A (x = 80.53, 1.5:1)



#	Part	Qty	Spec / use
1	top_plate (colour A)	1	PLA/PETG, top face down
2	frame (colour B)	1	PLA/PETG, accent colour
3	base (colour A)	1	PLA/PETG
4	bar_spine	1	PLA/PETG, 100 %
5	knob (colour B)	1	PLA/PETG, top face down, press fit
6	M3x8 countersunk ISO 10642	8	case screws
7	M3 heat-set insert 5.7x4.6	8	plate bosses
8	M7 nut + washer (KY-040 kit)	1	encoder panel mount
9	M2x4 self-tapping	4	1.9in display
10	M2x8 self-tapping	2	bar display + spine
11	M2x5 self-tapping	6	Pico 2 standoffs, LED sticks
12	WS2812B stick 8x 5050 (CJMCU)	2	base posts, LEDs up
13	Rubber feet 10x3	4	base recesses
14	MX switch + 1u keycap	12	plate mount
15	1N4148 diode	12	COL2ROW

NYXILAB MACROPAD / ROTARY EDITION

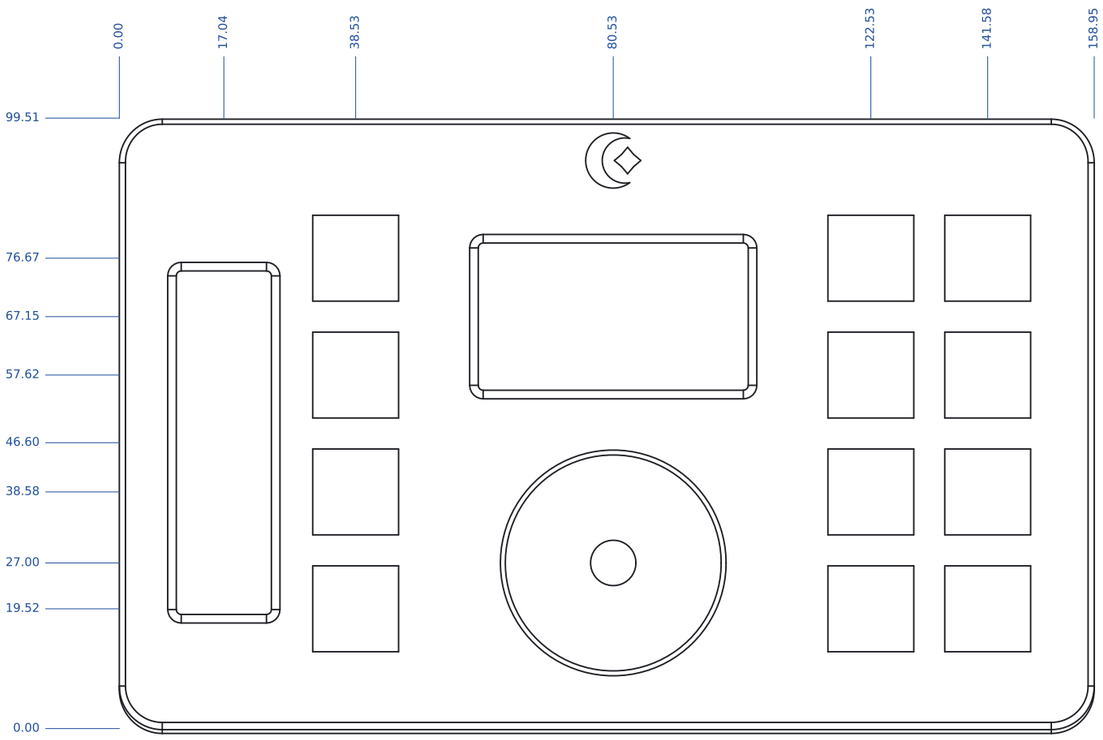
GENERAL ARRANGEMENT	SHEET 01 SCALE 1:1.5 / A-A 1:1
158.9 x 98.0 mm, 10 deg wedge, 3-layer stack	mm	2026-11-13

Generated from cad/macropad_cad (build123d). Do not scale the PDF - use the dimensions or the STEP files.

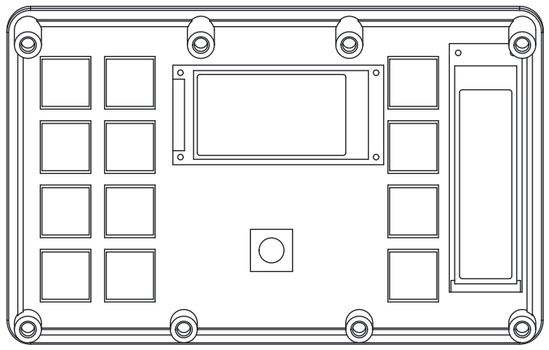
Features

MX cut-outs 12x 14.0 x 14.0 through, clip relief 15.0 x 15.6 from below, skin 1.5
Key pitch 19.05 | columns u = 38.53 / 122.53 / 141.58
Rows v = 19.52 / 38.58 / 57.62 / 76.67
1.9in window 44.02 x 24.00 centred (u 80.53, v 67.15), 1.4 x 45 bevel
2.25in window 15.50 x 55.99 centred (u 17.04, v 46.60), 1.4 x 45 bevel
Encoder mount at (u 80.53, v 27.00): D7.4 bushing hole through a 2.0 clamp section
  12.5 square key pocket from below, engraved ring D36.0 x 0.8 wide x 0.5 deep
Plate 5.0 thick, display skin 2.0, top edge chamfer 1.0
Slope length V = 99.51 (world depth 98.0 at 10 deg)

Ordinates from the front-left corner of the top surface (u right, v up-slope).



TOP (visible face)

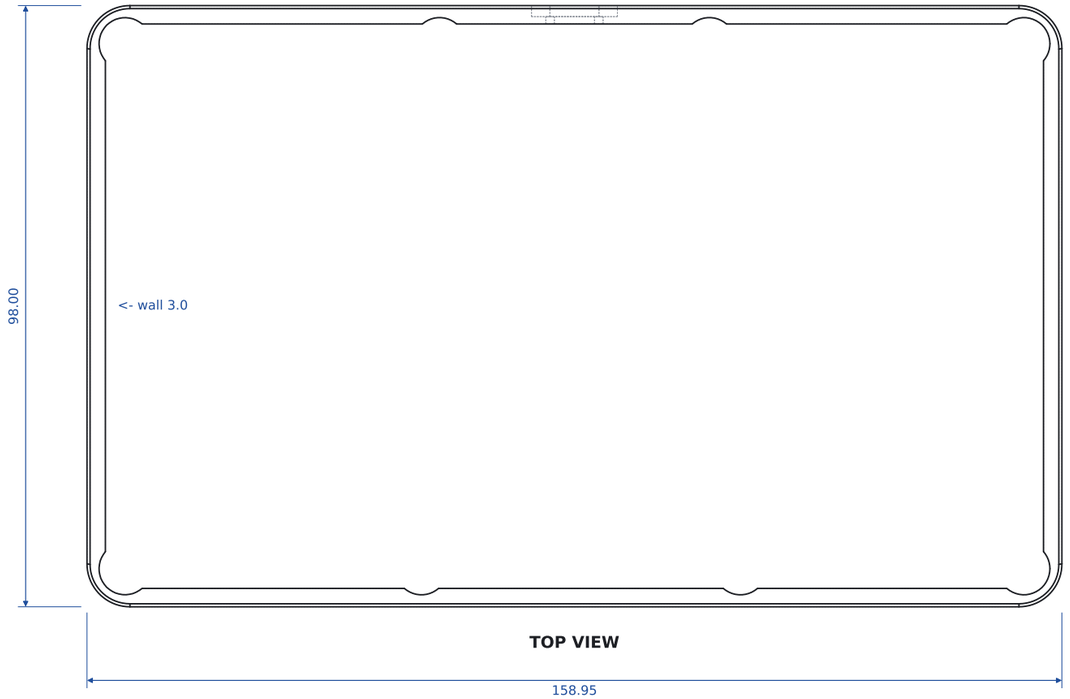


UNDERSIDE (from below, 1:1.8)

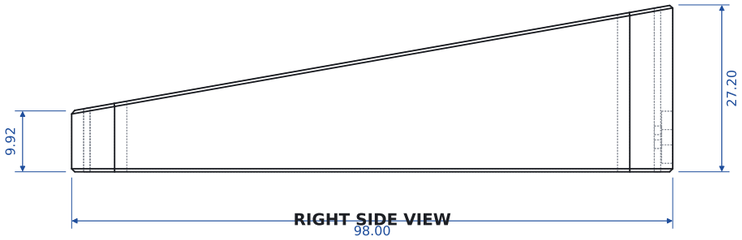
Bosses: world-vertical, M3 heat-set insert D4.0 x 6.2 at the lower end
Display pockets: glass + PCB steps, M2 pilot D1.7 blind (0.7 skin left)
Header relief grooves over the display pin rows (front-side joints)
Locating lip 1.6 x 2.5 drops inside the frame
Encoder body sits in the 12.5 square pocket: its M7 nut on top clamps 2.0 mm of plate

Generated from cad/macropad_cad (build123d). Do not scale the PDF - use the dimensions or the STEP files.

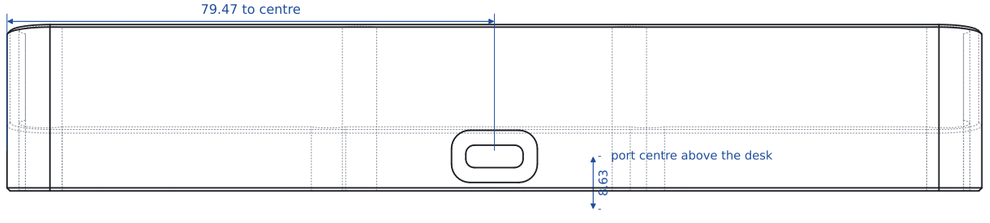
NYXILAB MACROPAD / ROTARY EDITION	
TOP PLATE - TRUE VIEW	SHEET 02 SCALE 1:1
Viewed normal to the tilted top surface (plate frame u, v)	mm | 2026-11-13



TOP VIEW



RIGHT SIDE VIEW



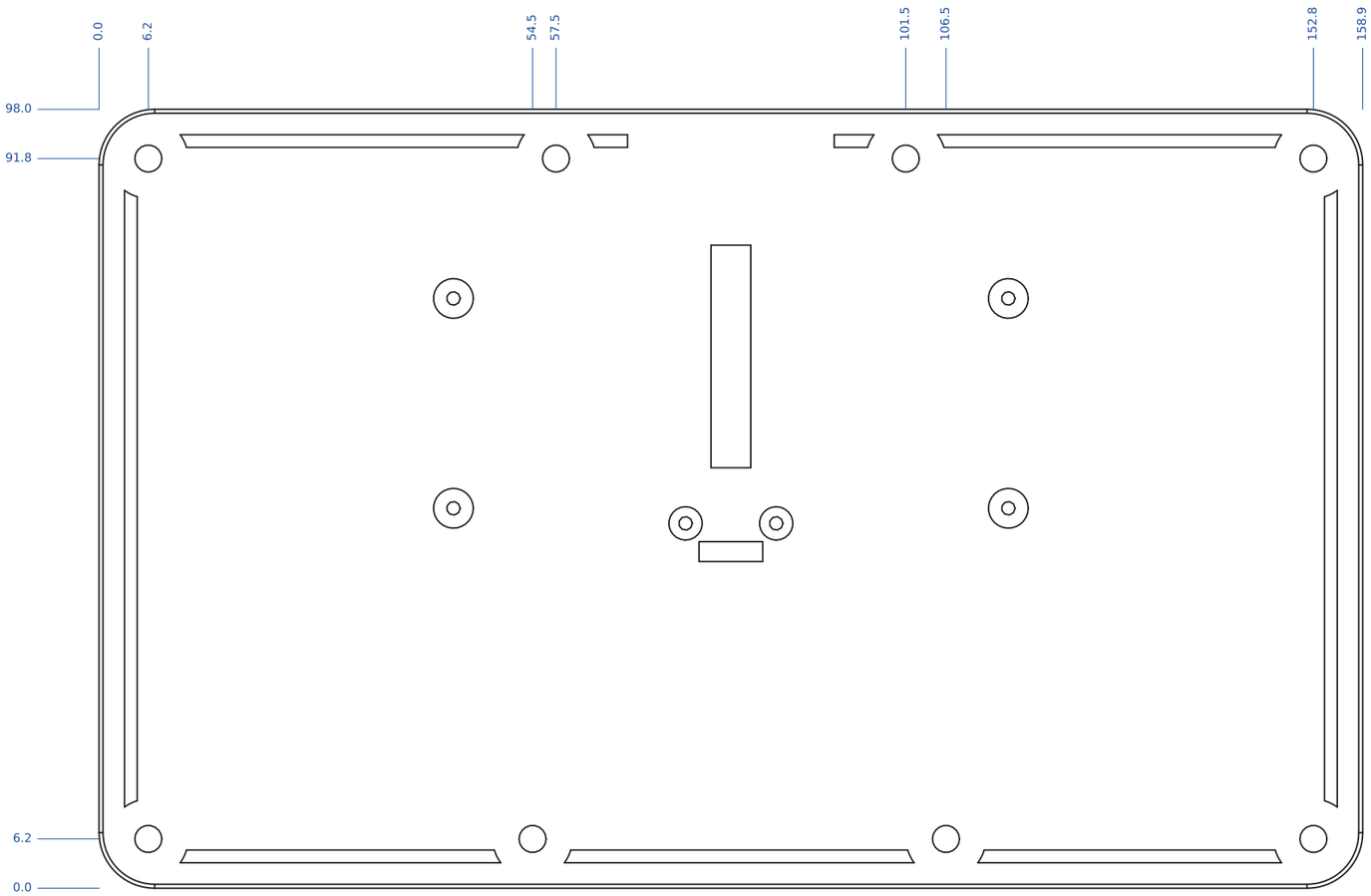
BACK VIEW (looking at the USB port)

USB-C port & details

Port centre x = 79.47 from the left edge, 8.63 above the desk
Thin wall 1.2 with receptacle opening 9.34 x 3.66 (R1.4)
Outer recess 14.0 x 8.5 R3.0, 1.8 deep, for the plug overmold
Receptacle face ends flush with the recess floor: any USB-C cable fits,
and the port captures the Pico's USB end (the clone has no holes there).
Boss saddles: 8x R4.25 vertical grooves locate the frame
0.5 mm chamfers at both seams form V-grooves between colours

NYXILAB MACROPAD / ROTARY EDITION	
FRAME (CENTRE LAYER)	SHEET 03 SCALE 1:1
Wedge wall ring, accent colour. USB-C port in the back wall.	mm | 2026-11-13

Generated from cad/macropad_cad (build123d). Do not scale the PDF - use the dimensions or the STEP files.



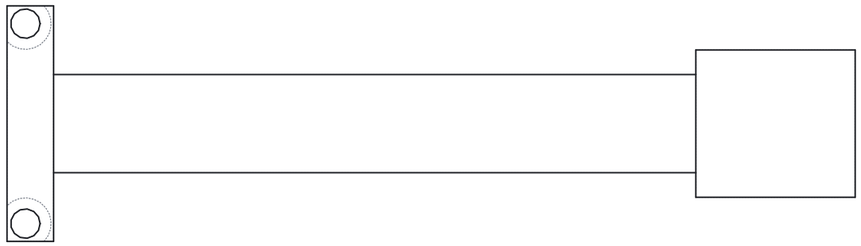
TOP VIEW 1.3:1 (inside face)

Features

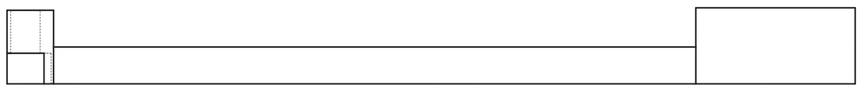
8x countersunk D3.4 / D6.8 x 90 deg (M3x8 ISO 10642)
Boss centres 6.2 from the edges; see ordinates
Pico standoffs D4.2 x 3.0, M2 pilot D1.7: (73.78, 45.90), (85.18, 45.90)
Pico centre rail + forward stop (USB end is held by the frame port)
No posts: the KY-040 hangs from the top plate on its M7 nut,
so the floor under it stays flat
LED stick posts 4x D5.0 x 3.0, M2 pilot D1.7 (2 sticks, LEDs up):
(114.37, 74.20), (114.38, 47.80),
(44.57, 47.80), (44.57, 74.20)
Feet: 4x D10.5 x 1.0 recesses, 16.0 from the edges
Locating lip 1.6 x 2.0 inside the frame (gaps at bosses & USB)

NYXILAB MACROPAD / ROTARY EDITION	
BASE (BOTTOM LAYER)	SHEET 04 SCALE 1.3:1
Pico 2 cradle, LED posts, M3 countersinks, feet	mm | 2026-11-13

Generated from cad/macropad_cad (build123d). Do not scale the PDF - use the dimensions or the STEP files.



SPINE - TOP 2:1



SPINE - SIDE 2:1 (bed face down)

Bar display spine

Held by the bar module's own two M2 holes (M2x8 through spine + PCB into the plate).
Arches 3.0 mm over the FPC / ZIF / parts on the module back,
pad presses the PCB 3-16 mm from the header end (0.2 mm preload).
Print flat side down, 100 % infill.

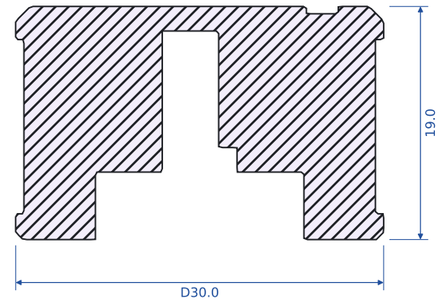
Fit-test coupons (print these first, ~1 h total)

fit_switch 3 MX holes 13.9 / 14.0 / 14.1 - pick the one your switches click into
fit_plate_d19 1.9in window, pockets and M2 pilots
fit_plate_bar 2.25in window, pockets and pilots
fit_plate_enc encoder mount: M7 bushing hole, square key pocket, nut clamp
fit_usb back wall + Pico cradle: plug a cable in, check alignment
knob print it early too: it should press onto the D-shaft firmly
fit_led base strip with the posts of one LED stick: check the hole pitch

If a fit is off, change the value in the edition's edition.py or common/cad/macropad_cad/params.py
and rebuild (make cad). Values marked MEASURE are the likely suspects.

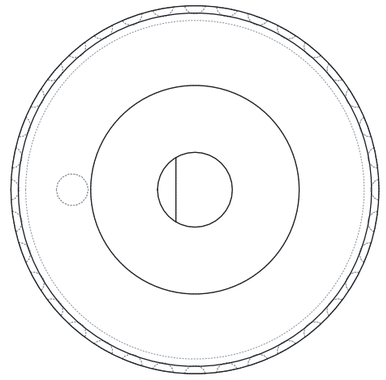
Knob (rotary edition)

D30 x 19, 36 flutes, 1.2 top chamfer, indicator dimple D2.6 at R10
D-bore D6.10, 4.60 across the flat; round below the flat so it only grips the D
Bore ends at the shaft tip (2.0 under the top): press on until it stops
Skirt recess D17.0 x 5.5 clears the M7 nut, washer and bushing thread
Rides 1.0 above the plate. Print top face down, 4 walls, 30 % infill, no supports
Too tight / loose on the shaft: Knob.bore_clear in rotary/cad/edition.py (0.10 now)



KNOB - SECTION THROUGH THE D-FLAT 2:1

(printed top face down, i.e. upside down from this view)



KNOB - FROM BELOW 2:1

NYXILAB MACROPAD / ROTARY EDITION

SPINE, KNOB & FIT TESTS	SHEET 05 SCALE 2:1
Bar-display clamp, the knob, coupons to print first	mm | 2026-11-13

Generated from cad/macropad_cad (build123d). Do not scale the PDF - use the dimensions or the STEP files.